THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

NORTH WESTERN ZONE FORM FOUR FORM PRE-NATIONAL EXAMINATION

031

TIME: 3HRS PHYSICS 18/9/2024

INSTRUCTION

1. This paper consists of sections A, B and C with a total of eleven (11) questions.
2. Answer all questions in section A and B and two (2) questions from section C.
3. Mathematical tables and non-programmable calculators may be used.
4. Cellular phones are not allowed in the examination room.
5. Write your Examination number / Names on every page of your answer booklet(s).
6. Where necessary the following constants may be used:
   (i) Acceleration due to gravity g=10m/sec$^2$
   (ii) Density of water = 1.0g/cm$^3$

**SECTION A (15 MARKS)**

**Answer all questions in this section**

1. For each of the items (i) – (ii), choose the correct answer among the given alternatives and write its letter besides the item number in the answer booklet provided.
   i. A radiation P is focused by a proper device on the bulb of a thermometer. Mercury in the thermometer shows a rapid increase. What does radiation P represent?
      A. infrared radiation
      B. visible light
      C. ultraviolet radiation
      D. X-rays
      E. Microwaves

ii. A body is acted upon by two unequal forces in opposite directions, but not in the same line. The effect is that:

A. the body will have only the rotational motion

B. the body will have only the translational motion

C. the body will have neither the rotational motion nor the translational motion

D. the body will have rotational as well as translational motion

E. the body will have no rotational motion

iii. A body moved upward a distance of 20m. What is the time taken to reach the maximum height?

A. 2sB. 5sC. 10s

D. 15sE. 11s

iv. Which of the following is the way through which electrical appliances in a house are connected?

A. Series

B. Parallel

C. Either in series or parallel

D. Both in series and parallel

E. None

v. A magnified erect image is obtained when an object is positioned in front of a converging lens. What is the distance of the object from the lens?

A. Greater than three focal lengths

B. Equal to two focal lengths

C. Between one and two focal lengths

D. Equal to one focal length

E. Less than one focal length

vi. Why diffusion occurs more quickly in a gas than in a liquid?

A. The liquid contains a layer on its surface

B. The gas contains semi-permeable membrane

C. The gas molecules are small in size compared to the liquid molecules

D. The adhesion is large than cohesion in gas compared to that in liquid.

E. The speed of molecules in gas is greater than in liquid.

vii. A magnetic needle, free to turn in a vertical plane, is suspended first at the earth's south magnetic pole and then at a point on the magnetic equator. The respective angles between the needle and horizontal are:

A. $0^o$ and $0^o$

B. $70^o$ and $70^o$

C. $0^o$ and $45^o$

D. $0^o$ and $90^o$

E. $90^o$ and $0^o$

viii. A ray of light incident on a lens parallel to its principal axis, after refraction passes through or appears to come from:

A. its first focus

B. its optical entre

C. its second focus

D. the Centre of curvature of its second surface

E. the pole

ix. A typist uses a new carbon paper under her top typing paper for making a copy of a letter. When she holds the carbon paper close to a plane mirror, she can read the letter. This is because the mirror;

A. forms an image the same size as object

B. produces an inverted image

C. produces a laterally image

D. forms a virtual image

E. forms an image behind the mirror

x. What happen when a ray of light from air enters a denser medium?

A. Bends away from the normal

B. Bends towards the normal

C. Goes un deviated

D. Is reflected back

E. Is diffracted

2. Match the items in List A with the responses in List B by writing the letter of the correct response in your answer sheet/booklet.

| LIST A | LIST B |
|---|---|
| I. Electromagnetic radiations which are used to increase the rate of growth of chickens.<br>II. Electromagnetic radiations which are used by the Global Positioning Satellite (GPS) to precisely pin point the location of the targets.<br>III. Electromagnetic radiations which are used for satellite communications or telecommunications.<br>IV. Electromagnetic radiations which are used as painkillers.<br>V. Electromagnetic radiations which are used in CD and DVD players to read the content of the disks. | A. Radio waves<br>B. Microwaves<br>C. Infra-red<br>D. Visible light<br>E. Ultra-violet<br>F. X-rays<br>G. Gamma ray |

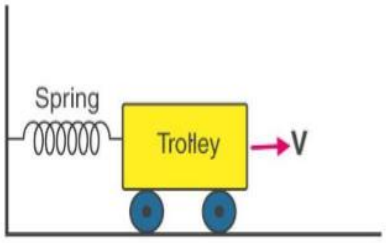
**SECTION B: (60 MARKS)**

**Answer all questions in this section**

3. (a) (i)The sinking of the wheels of a lorry into soft sand may be prevented by letting off some air from the tires. Explain this observation. **[4marks]**

   (ii) Why atmospheric pressure never breaks glass windows of a house? **[4marks]**

   (b) A spring is kept compressed by a small trolley of mass 0.5 kg lying on a smooth horizontal surface as shown.




   When the trolley is released, it is found to move at a speed of v=2m/s. What potential energy did the spring possess when compressed? **[2marks]**

4. (a) Describe an experiment to show that sound can travel in a water. **[4marks]**

(b) Two men stand a distance apart besides a long metal rail on a sill day. One man places his ear against the rail while the other gives the rail a sharp knock with a hammer. Two sounds separated by a time interval of 0.5s, are heard by the first man. If the speed of sound in air is $330ms^{-1}$, and that in the metal rail is $5280ms^{-1}$, find the distance between the men. **[6marks]**

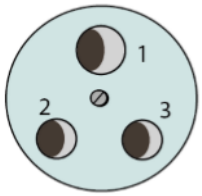
5. (a) (i)State any two examples where the law of conservation of energy is observed. **[2marks]**

   (ii) Distinguish between elastic collision and inelastic collision. **[4marks]**

   (b) An arrow of mass 100g is shot into a block of wood of mass 400g lying at rest on the smooth surface of an ice rink. If at the moment of impact, the arrow is travelling at15m/s, calculate the common velocity after the impact. **[4marks]**

6. (a)It is dangerous to connect the switch in the neutral wire. Explain your answer. **[2marks]**

   (b) The diagram in Fig. shows a three-pin socket marked as 1, 2 and 3.

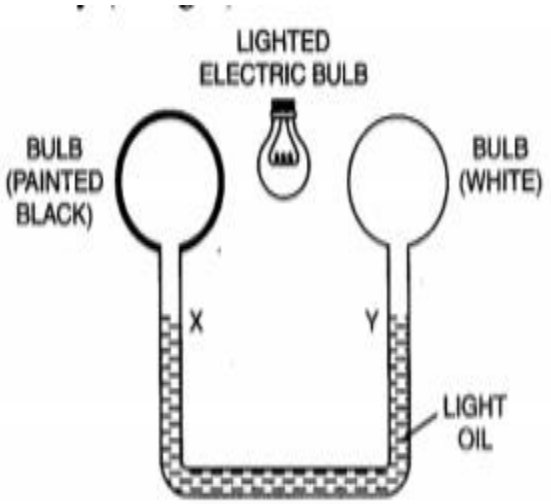



   (i) Identify and write live (L), neutral (N) and earth (E) against the correct number. **[3marks]**

   (ii) To which part of the appliance is the terminal 1 connected? **[3marks]**

   (iii) To which wire joined to 2 or 3, is the fuse connected? **[2marks]**

7. (a) The diagram in the figure below shows two air filled bulbs connected by a U-tube partly filled with a low density (or light) oil.




(i) Describe how the levels of oil in the limbs X and Y change after the electric bulb is lighted. **[6marks]**

(ii) Account for your observation. **[2marks]**

(b) It is difficult to cook vegetables on hills and mountains. Explain the reason. **[2marks]**

8. (a)(i) Ice cream appears colder to the mouth than water at 0oC. Give reasons. **[2marks]**

(ii) Explain why water is a preferable liquid for quenching thirst? **[4marks]**

(b) 45 g of water at 50oC in a beaker is cooled when 50 g of copper at 18oC is added to it. The contents are stirred till a final constant temperature is reached. Calculate the final temperature. The specific heat capacity of copper is 0.39 Jg-1K -1 and that of water is 4.2 Jg-1K -1 . State the assumptions used. **[4marks]**

## SECTION C (25 MARKS)

**Answer any two (2) questions form this section**

9. (a) The relative density of a substance is 0.65. Explain the meaning of this statement. **[2.5marks]**

(b) An ordinary hydrometer of mass 28g floats with 3cm of its stem out of water. The cross-section area of the stem is $0.75cm^2$ . Find;

(i) the total volume of the hydrometer. **[3marks]**

(ii) the length of stem above the surface when it floats in a liquid of relative density 1.4. **[3marks]**

(c) A piece of cork of volume 100cm3 is floating on water. If the density of cork is 0.25g/cm3 . Calculate;

(i) the volume of cork immersed in water. **[1mark]**

(ii) the force needed to immerse the cork completely. **[3marks]**

10. (a) Why do recording studios and concert auditoriums sometimes have their ceilings and walls covered with soft perforated boards? **[2.5marks]**

    (b)The frequency of the 3rd overtone (4th harmonic) produced by an open pipe is $840Hz$ . Given that the velocity of sound in air is 330m/s, calculate;

    (i) Length of the open pipe **[3marks]**

    (ii) Fundamental frequency **[3marks]**

    (c) A stretched string is in unison with a turning fork of frequency 384Hz. If the tension in the string is increased by 2% what is the frequency of the beats heard when the fork and the string are now sounded together? **[4marks]**

11. (a) Give any five differences between Human eye and lens Camera. **[5marks]**

    (b) What is the role played by the shutter in a Lens camera? **[2.5marks]**

    (c)A compound microscope has an objective lens of focal length 2cm and eye piece of focal length of 6cm. An object is placed 2.4cm from the objective lens. If the distance the objective lens and the eye piece lens is 17cm. Find;

    (i) The distance of the final image from the eye piece **[2marks]**

    (ii) The linear magnification. **[3marks]**